PROJECTILE MOTION I

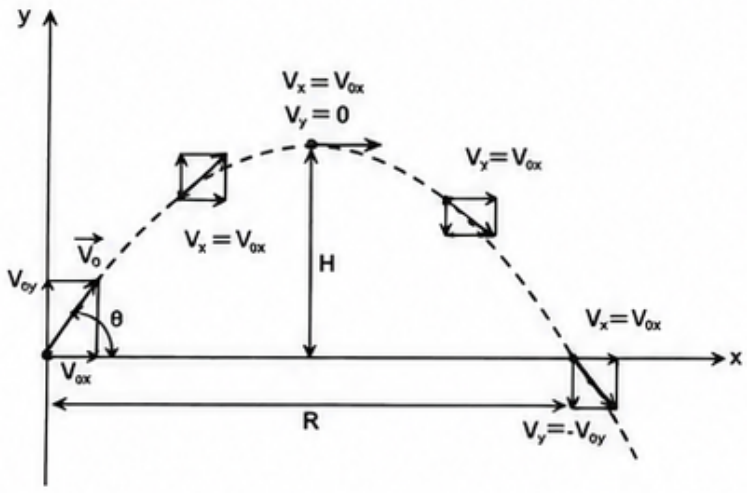
LEARNING GOALS

Students will:

- Describe the motion of an object launched at arbitrary angles through the air.
- Describe the horizontal and vertical motions of a projectile.
- Solve projectiie motion problems.

CONCEPT PROBLEMS

1. What is the shape of the path followed by the projectile? A parabola
2. If a projectile is launched horizontally, what is its initial vertical velocity? 0 m/s
3. For a projectile that lands at the same level it's launched at, predict:
   a. What launch angle will give a projectile maximum horizontal distance?
      45° above the horizantal
   b. What other angle gives you the same distance as 30°? As 55°?
      For 30°, 60° gives the same distonce
      For 55°, 35°, gives the same distance
   c. What is the rule for finding two angles that give the same distance?
      Any 2 angles that add up to 90° give the same distance
4. The diagram below shows the path of a projectile as at flies through the air.



   a. Describe the horizontal and vertical motion of a projectile.
      - The horizontal motion is uniform
      - The vertical motion has uniform acceleration with a changing velocity
   b. Based on your answer to a, what is the **only** equation you will always use to analyze the **horizontal** motion of a projectile?
      $\vec{\Delta d}x = \vec{V}x\,\Delta t$
   c. Focus on the overall velocity. (The central arrows are the overall velocity). Does it ever become zero? Why or why not?
      No, the overall velocity is never zero because there is always a constant horizontal velocity even when the vertical velocity is zero at the vertex.
   d. Consider how the overall velocity is oriented to the path of the projectile. What word do we use to describe this orientation?
      The velocity is always tangent to the path.

IMPORTANT RULE

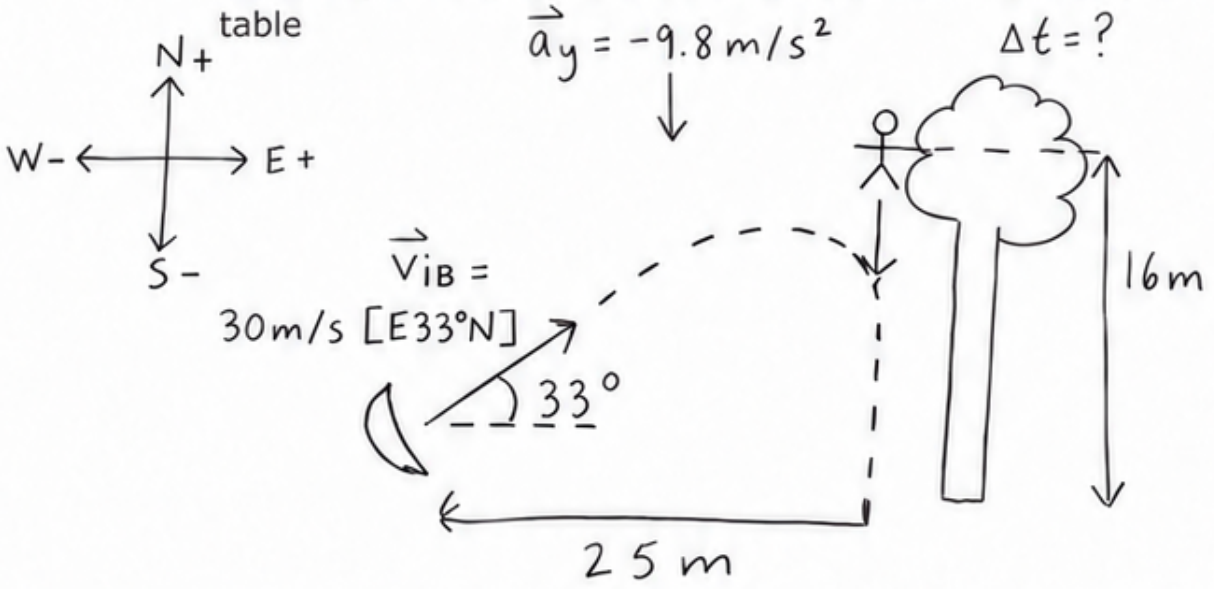
The horizontal part of the motion is independent from the vertical motion. The two parts can be solved separately. HOWEVER, the TIME to complete the horizontal motion is the same as the TIME to complete the vertical motion.

Recall that the horizontal motion of a projectile is uniform motion and the vertical component is under the influence of gravity so projectiles always have a horizontal acceleration ($\vec{a}_x$) of $0\ m/s^2$ and a vertical acceleration ($\vec{a}_y$) of $-9.81\ m/s^2$

PROBLEM SOLVING

In the shoot the monkey scenario, if the cannon fired the banana with the initial velocity of 30.0 m/s at an angle of 33.0° above the horizontal and the tree is 16.0 m tall and 25 m away from the cannon, how far does the monkey fall before it catches the banana? Answer this question by breaking it down into the steps below.

- Make a quick labelled sketch of the situation. Define positive x and y. Then fill in the table



	x	y
$\Delta\vec{d}$	25 m	11 m
$\vec{v}_1$	+25.2 m/s	+16 m/s
$\vec{v}_2$	+25.2 m/s	6.3 m/s
$\vec{a}$	$0\ m/s^2$	$-9.8\ m/s^2$
Δt	0.992 s	0.992 s

- Why is it impossible to calculate anything with the vertical component right now?

because we only have 2 known value, whizh would leave 2 unknown values in the equation, which is unsolvable.

- How can you use the x-component to help? Do it. Then go back and solve the vertical.

$$\vec{\Delta d}_x = \frac{\vec{v}_{2x} + \vec{v}_{ix}}{2}\Delta t$$

$$\Delta t = \frac{2\vec{\Delta d}_x}{\vec{v}_{2x} + \vec{v}_{ix}}$$

$$\Delta t = \frac{2 \times 25}{25.2 + 25.2}$$

$$\Delta t = 0.99\ s$$

$$\vec{\Delta d}_y = \vec{v}_{1y}\Delta t + \frac{1}{2}\vec{a}_y \Delta t^2$$

$$\vec{\Delta d}_y = 16 \times 0.992 + \frac{1}{2}(-9.8)(0.992)^2$$

$$\vec{\Delta d}_y = 11\ m$$

$$\vec{v}_{2y} = \vec{v}_{1y} + \vec{a}_y \Delta t$$

$$\vec{v}_{2y} = 16 + (-9.8)(0.992)$$

$$\vec{v}_{2y} = 6.3\ m/s$$

$$\Delta d_{ym} = \vec{v}_{im}\Delta t + \frac{1}{2}\vec{a}_m \Delta t^2$$

$$\Delta d_{ym} = 0 \times 0.992 + \frac{1}{2}(-9.8)(0.992)^2$$

$$\vec{\Delta d}_{ym} = -4.8\ m$$

EXAMPLE

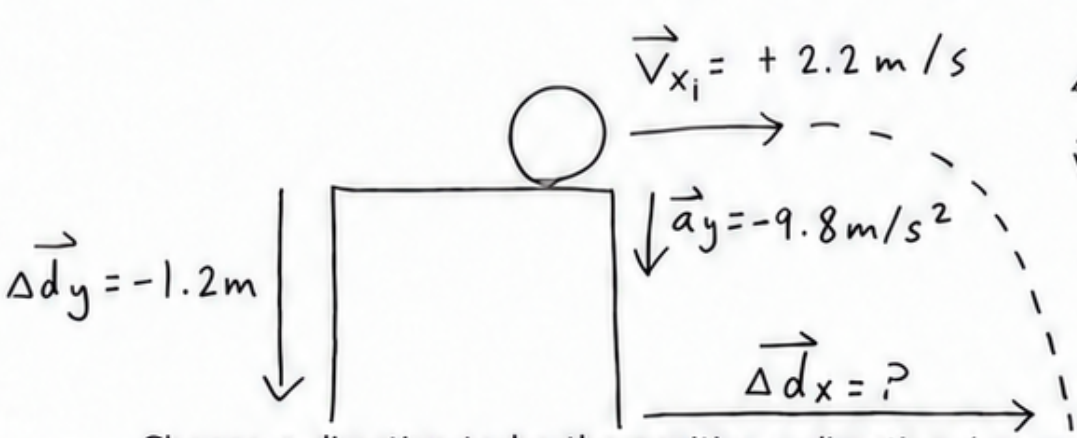
A ball is rolled at 2.2 m/s along a counter top that is 1.2 m above the floor. It rolls over the edge of the counter top.

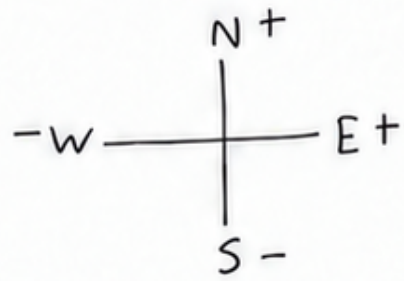
a) How far horizontally does it travel before hitting the floor?

b) What is the velocity of the ball when it hits the floor?

SPLIT THE GIVENS INTO HORIZONTAL AND VERTICAL QUANTITIES. Time: is the same for both x and y.

Draw a fully labelled diagram showing the situation. **All labels should indicate either x- or y-**





Choose a direction to be the positive y-direction (up or down). Label it and make sure all your positives and negatives are correct.

a) Horizontal motion is uniform. Write the equation for $\Delta\vec{d}_x$: $\vec{V}x \cdot \Delta t$

But you need to find the time first. Use the vertical motion:

$\Delta\vec{d}_y = \vec{V}_{iy}\cancel{^{0}\Delta t} + \frac{1}{2}\vec{a}_y \Delta t^2$ $\quad 0.245 = \Delta t^2$

$-1.2 = \frac{1}{2}(-9.81)\Delta t^2$ $\quad \sqrt{0.245} = \Delta t$

$-1.2 = -4.905\,\Delta t^2$ $\quad 0.49s = \Delta t$

$\Delta d x = \vec{V}x\ \Delta t$

$\Delta\vec{d}x = 2.2 \times 0.49$

$\Delta\vec{d}x = 1.09\ m$

b) The final velocity has two components. The horizontal velocity is $2.2\ m/s$ (constant).

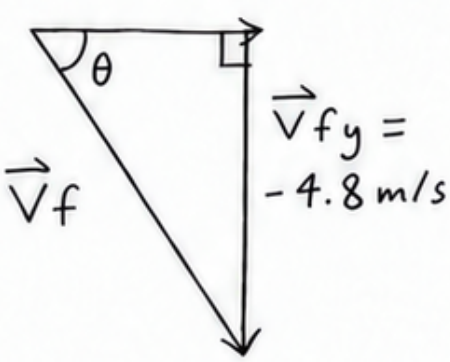
Now find the final vertical velocity:

$\vec{V}fy = \vec{V}_{iy} + \vec{a}_y \Delta t$ $\quad \vec{V}fy = -4.8\ m/s$

$\vec{V}fy = 0 + (-9.8)(0.49)$

$\vec{V}fx = +2.2\ m/s$

Label the diagram and then find the overall final velocity:



$\vec{V}f = \sqrt{\vec{V}fx^2 + Vfy^2}$ $\quad \theta = \tan^{-1}\left(\frac{\vec{V}fy}{\vec{V}fx}\right)$

$Vf = \sqrt{(2.2)^2 + (-4.8)^2}$ $\quad \theta = \tan^{-1}\left(\frac{4.8}{2.2}\right)$

$Vf = 5.3\ m/s$ $\quad \theta = 65°$ south of east

$\therefore \vec{V}f = 5.3\ m/s\ [E\ 65°S]$

Answers: 0.495 s; 1.09 m
$\vec{v}_{fy} = -4.85$ m/s, $\vec{v}_{fx} = 2.2$ m/s,
$\vec{v}_f = 5.33$ m/s [66° below horizontal]

homework: Pg.40 Q#1-4

ADDITIONAL RESOURCES

Readings

- Nelson Physics 12, pg.36-40, Work through Tutorial 1 – Sample Problems
- The Physics Classroom
  - What is a Projectile?
  - Characteristics of a Projectile's Trajectory

Videos

- Khan Academy
  - Deriving Max Projectile Displacement Given Time
  - Projectile Height Given Time
  - Total Displacement for Projectile
  - Total Final Velocity for Projectile

Activities

- Complete the activity on pg.1-2 of this note